

An Autonomous Context-Aware Framework for Deterring Human Curiosity Against USB Baiting Attacks in Enterprise IT Environments

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⋮ **ABSTRACT** As organizational network perimeters become increasingly fortified...

As organizational network perimeters become increasingly fortified, cyber adversaries progressively shift their focus toward human-centric vulnerabilities, utilizing physical media in the form of social engineering vectors. Among these, USB-based baiting attacks remain highly effective, exploiting human curiosity and cognitive biases to bypass conventional digital defenses. Traditional security approaches predominantly rely on post-incident detection or rigid administrative policies, which often fail to account for suboptimal human decision-making under psychological triggers. To address this critical security gap, this research proposes a novel, multi-layered endpoint defense framework designed to mitigate USB-based baiting attacks by harmonizing hardware isolation with behavioral economics. The proposed architecture integrates two primary defensive layers: a hardware-level isolation mechanism that sandboxes untrusted USB devices upon insertion to prevent immediate architectural compromise, and a proactive behavioral layer that deploys real-time cognitive "nudges" to alter user interaction pathways without restricting autonomy. The methodology encompasses the design of the isolation framework, the development of context-aware behavioral nudges rooted in dual-process theory, and an empirical evaluation to assess both technical resilience and psychological deterrence. Ultimately, this research aims to bridge the gap between technical endpoint security and behavioral cybersecurity, delivering a robust, human-centric defense model that significantly reduces the success rate of physical media social engineering attacks in enterprise environments.

⋮ **INDEX TERMS** Behavioral Cybersecurity, Cognitive Nudges, Endpoint Defense, Hardware Isolation, USB Baiting.

I. INTRODUCTION

THE modern enterprise computing landscape has witnessed an exponential proliferation of sophisticated cybersecurity frameworks, ranging from next-generation firewalls to artificial intelligence-driven endpoint detection and response systems. Despite these massive technological investments, organizations remain highly vulnerable to catastrophic security breaches due to a persistent and complex vulnerability: the human element. Cybercriminals increasingly bypass rigid perimeter defenses not by breaking encryption algorithms, but by deploying social engineering strategies that exploit human cognitive biases and psychological triggers.

A. BACKGROUND OF THE STUDY

A preliminary scan of recent academic literature and industry reports indicates that current organizational approaches to mitigating USB baiting attacks are severely constrained by their dependency on static or reactive methodologies. On one hand, human-centric mitigation relies almost entirely on periodic Security Awareness Training (SAT), which attempts to educate employees on the theoretical risks of unauthorized media usage. However, behavioral psychology and empirical cybersecurity studies consistently demonstrate that theoretical knowledge often fails to translate into secure behavior in real-time environments, as intense curiosity or urgent urgency frequently overrides logical risk assessment [1].

On the other hand, technical infrastructure controls depend

on traditional Endpoint Protection Platforms (EPP) or rigid Host Intrusion Prevention Systems (HIPS). These tools typically operate under a binary management paradigm—either completely blocking all external peripheral interfaces, which introduces immense operational friction and degrades workplace productivity, or allowing them under standard antivirus scanning, which creates a dangerous security lag because signature-based detection cannot preempt zero-day payloads or specialized hardware-level attacks (such as Rubber Ducky or BadUSB devices) at the exact moment of physical insertion [2].

B. PROBLEM STATEMENT

Modern enterprise endpoint security paradigms remain acutely vulnerable to human-centric social engineering vectors, specifically physical Universal Serial Bus (USB) baiting attacks, which systematically exploit human curiosity to bypass sophisticated perimeter defenses. While contemporary research heavily prioritizes reactive post-execution software security—such as signature-based malware detection and sandboxing—this approach creates a critical security lag, allowing malicious payloads to execute immediately upon the operating system mounting the peripheral file system. Furthermore, existing host-level logging mechanisms suffer from a profound analytical and structural vulnerability: they record technical event metadata (e.g., system timestamps and hardware IDs) but completely fail to establish non-repudiation or verify the physical identity of the individual who inserted the device. This empirical gap creates a critical evidentiary failure in digital forensics, as security administrators cannot definitively determine whether a security breach was initiated by an inside threat, a negligent employee, or an unauthorized physical intruder exploiting an unattended workstation.

Consequently, conventional defensive systems leave organizations exposed to zero-day hardware exploits and lack the capability to execute real-time, pre-execution containment at the hardware-software boundary. There is a demonstrable omission in cybersecurity literature regarding an integrated architecture that synthesizes low-level kernel driver isolation, real-time psychological deterrence, and automated hardware-triggered forensic asset capture. Without an automated framework that concurrently freezes unauthorized peripheral drivers, activates local camera interfaces to secure immutable physical evidence, and deploys cognitive nudges at the user interface layer, enterprise endpoints remain structurally incapable of disrupting user curiosity or ensuring absolute accountability during physical social engineering events. A critical analysis of current enterprise defense mechanisms reveals four distinct technological, procedural, and evidentiary gaps:

Standard Endpoint Protection Platforms (EPP) and traditional antivirus engines operate under a post-insertion detection model. They only scan or neutralize threats after the operating system mounts the file system or when a file script attempts to execute. This creates a highly dangerous secu-

rity lag. If the unauthorized physical media contains zero-day exploits, advanced payload obfuscation, or specialized hardware-level threat emulators (such as BadUSB or Rubber Ducky configurations), the initial host compromise occurs within milliseconds of physical insertion, rendering reactive defense software ineffective at the critical hardware-software boundary [3].

1) The Behavioral-Compliance Gap

Existing Information Technology Governance frameworks assume static, rule-abiding behavior from personnel based on periodic Security Awareness Training (SAT). However, behavioral psychology and empirical security data demonstrate that cognitive biases—specifically curiosity combined with urgent contextual labeling—frequently lead employees to bypass theoretical security training in real-time situations. Traditional systems provide no real-time psychological intervention or immediate cognitive friction to disrupt the human impulse to explore unauthorized media at the exact point of risk [4].

2) The Evidentiary and Identity Non-Repudiation Gap

A profound vulnerability in corporate IT auditing is the disconnection between digital system logs and the physical environment. While operating systems record that a peripheral device was connected, they remain fundamentally blind to *who* initiated the action. Traditional defenses cannot distinguish between an authorized employee committing an error, a malicious insider acting intentionally, or an external intruder exploiting a temporarily unlocked terminal. This lack of automated physical-to-digital evidence binding invalidates forensic non-repudiation, making it impossible to establish definitive legal and operational accountability post-incident [5].

3) Operational Binary Inflexibility

Current Universal Serial Bus (USB) port management tools operate on rigid, binary administrative rules—either completely blocking all external hardware peripheral interfaces or completely allowing them under standard user profiles. Total interface blocking introduces massive operational friction, impedes legitimate business workflows, and harms administrative productivity. Conversely, allowing unverified device interactions under standard profiles leaves the entire enterprise infrastructure exposed. Current IT controls lack the situational intelligence required to dynamically adjust host security restrictions based on the changing context of the user and the environment.

C. PURPOSE OF THE STUDY

The primary purpose of this study is to bridge the critical technological and evidentiary gap between digital endpoint vulnerabilities and human behavior during physical social engineering events. Specifically, this research aims to design, develop, and evaluate an autonomous, context-aware defensive framework that systematically neutralizes Universal

Serial Bus (USB) baiting threats at the immediate hardware-software boundary while establishing absolute physical accountability.

By engineering a unique multi-layered architecture, this study shifts enterprise security from a reactive, software-centric scanning model to a proactive, hardware-triggered containment paradigm. Mechanistically, the framework is designed to achieve two overarching outcomes: first, to enforce immediate, low-level physical interface isolation (driver-level freezing) to halt potential malicious code execution before file system mounting occurs; and second, to deploy localized forensic camera activation to capture real-time visual evidence of the physical initiator alongside cognitive user-interface nudges. Ultimately, this research serves an institutional and psychological purpose. It aims to transform corporate workstations from passive targets into active compliance enforcement points. By introducing real-time behavioral friction combined with automated visual non-repudiation, the study seeks to understand how immediate awareness of physical-digital auditing alters human decision-making, effectively reducing organizational susceptibility to insider threats and curiosity-driven security breaches while maintaining an optimized operational workflow.

D. RESEARCH QUESTIONS

To systematically investigate the technical viability and behavioral efficacy of the proposed framework, this study addresses the following central overarching research question: *How can a context-aware defense architecture integrating low-level interface isolation, automated forensic capture, and real-time cognitive nudges mitigate enterprise vulnerability to physical USB baiting attacks without introducing unmanageable operational friction?* To resolve this primary inquiry, the research addresses the following specific, empirical sub-questions:

following specific, empirical sub-questions:

- 1) *Can a low-level hardware filter driver successfully...*
- 2) *Can a low-level hardware filter driver successfully intercept physical peripheral connection events and execute driver-level freezing before the host operating system mounts the file system, and what is the resulting optimization in containment latency?*
- 3) *How efficiently can an automated, hardware-triggered forensic asset capture protocol activate the local camera interface upon unauthorized device detection to secure immutable physical evidence and achieve digital non-repudiation?*
- 4) *To what extent does the concurrent deployment of real-time cognitive nudges and automated visual accountability notifications at the user interface layer reduce user susceptibility to USB baiting exploits within an enterprise cohort?*
- 5) *How effectively can a dynamic policy orchestration engine utilize real-time situational metadata to adaptively enforce peripheral security restrictions without*

causing statistically significant workflow degradation for authorized hardware interactions?

E. RESEARCH HYPOTHESES

To provide analytical thresholds and guide the empirical verification of the multi-layered endpoint security system, the following performance and behavioral hypotheses are formulated:

- H_1 : Low-level hardware filter driver interception demonstrates a statistically superior speed profile in isolating unauthorized interfaces compared to traditional reactive signature-scanning engines.
- H_2 : The deployment of live cognitive nudges integrated with dynamic visual auditing notifications creates significantly higher policy compliance across an operational enterprise cohort than passive compliance warning screens.
- H_3 : Instantaneous hardware-triggered camera activation creates immutable forensic cross-binding, improving visual non-repudiation tracking speed over standard logical account tracking.
- H_4 : Situational context-aware configuration scoring optimization yields lower structural productivity loss for enterprise users compared to uniform rigid administrative interface deactivation rules.

F. RESEARCH OBJECTIVES

The overarching goal of this research is to develop and quantitatively evaluate a proactive, forensic-enabled security framework that mitigates human-induced vulnerabilities and establishes absolute accountability at the physical-host interface layer. The specific research objectives established to accomplish this goal are as follows:

- 1) **Modeling Contextual Risk Profiles:** To identify, extract, and model the critical environmental metadata and situational cues—such as user organizational roles, login session risk scores, and device insertion context—that determine the dynamic risk profile of a physical hardware interaction.
- 2) **Designing Cognitive Deterrence Interventions:** To design and implement a lightweight, host-level behavioral deterrence engine capable of deploying real-time, context-specific psychological nudges to actively discourage unauthorized user interaction with untrusted media.
- 3) **Engineering Pre-Execution Interface Isolation:** To engineer an automated IT governance control layer that dynamically orchestrates proactive endpoint containment, specifically executing temporary physical interface isolation or driver-level freezing prior to file system mounting or payload execution.
- 4) **Integrating Automated Forensic Asset Capture:** To develop and embed an immediate, hardware-triggered forensic capture protocol that activates the host's localized camera interface upon unauthorized peripheral

attachment, securing instant visual evidence to achieve immutable physical-to-digital non-repudiation.

- 5) **Empirical Validation and Performance Benchmarking:** To construct an experimental, simulated corporate IT environment to quantitatively validate the integrated framework's performance against standard benchmark metrics, including attack deterrence success rates, interface isolation latency, forensic trigger velocity, and operational workflow friction thresholds.

G. SIGNIFICANCE OF THE STUDY

The significance of this research lies in its potential to redefine endpoint infrastructure security by establishing a novel, multi-dimensional nexus between behavioral psychology, automated Information Technology governance, and real-time digital forensics. As human-centric cyberattacks continue to effortlessly bypass sophisticated perimeter software, this study provides substantial academic, practical, and forensic contributions to the field of corporate technology management.

1) Theoretical and Academic Contributions

From an academic perspective, this research addresses a critical gap in the literature regarding the "human element" in cybersecurity. Although existing studies heavily separate technical endpoint protection from behavioral awareness, this study introduces an integrated model in which behavioral deterrence is directly embedded into the logic of the host-level system. By formalizing how situational metadata (such as organizational roles and login risk scores) can be translated into proactive defensive orchestration, this research provides a theoretical foundation for future studies in behavioral cybersecurity and human-centric threat mitigation. Furthermore, it advances choice architecture theory by exploring how the immediate perception of active physical monitoring influences cognitive biases and curiosity-driven policy compliance under risk-inducing conditions.

2) Practical and Operational Contributions for Enterprise IT

Practically, the proposed framework addresses the critical operational limitations of traditional IT controls. Rather than forcing organizations to choose between complete operational paralysis (total USB port blocking) or severe security exposures (unrestricted access), this study introduces a dynamic, context-aware alternative. IT administrators gain an automated governance layer capable of executing micro-interventions—such as deploying real-time cognitive nudges or temporary driver-level freezes—exactly when and where risk peaks. This dramatically minimizes human-induced data breaches while preserving user productivity and daily business continuity.

3) Institutional Forensic Accountability and Non-Repudiation

A major contribution of this study is the technical resolution of the identity attribution failure inherent to standard operating system audit trails. By introducing an automated,

hardware-triggered forensic asset capture mechanism, this research closes the critical gap between digital events and physical actors. When an unauthorized peripheral interaction occurs, the framework establishes instantaneous physical-to-digital evidence binding by leveraging the endpoint's localized camera interface. This provides corporate security teams with incontrovertible visual verification, ensuring absolute non-repudiation, accelerating internal incident triage, and offering an immutable forensic foundation required for regulatory investigations.

4) Regulatory Compliance and Risk Management

Finally, this research carries vital significance for enterprise risk management and regulatory compliance. Modern corporate environments must adhere to strict data protection regulations (such as ISO/IEC 27001, NIST SP 800-53, and GDPR) which demand comprehensive controls over physical access and data exfiltration vectors. By providing verifiable, automated logging of unauthorized peripheral events alongside proactive containment metrics and supporting physical evidence, the framework enhances institutional security auditing, significantly reduces corporate liability, and strengthens the overall resilience of critical digital assets against social engineering attacks.

5) Academic Gaps Addressed

To clearly differentiate this study from contemporary literature, it is essential to highlight how the proposed framework extends, refines, and improves upon the methodologies implemented by previous researchers. Current state-of-the-art defenses against social engineering and USB baiting generally fall into separate domains: static human education or reactive software-level mitigation. This research bridges this gap by introducing four distinct evolutionary improvements over existing works:

- **Shifting from Post-Execution Detection to Pre-Execution Deterrence:** The vast majority of existing endpoint security tools focus exclusively on analyzing the payload *after* the USB device is mounted by the operating system. This architectural logic creates a dangerous security lag. In contrast, this framework introduces a proactive defense mechanism that operates at the physical insertion layer, executing contextual verification and behavioral intervention before the file system is mounted.
- **Transitioning from Static Compliance to Dynamic, Context-Aware Governance:** Traditional IT governance relies heavily on rigid, binary policies—such as complete port blocking or static access control lists—which introduce high operational friction. This research replaces these static methods with a dynamic orchestration layer, interpreting situational metadata (e.g., user organizational roles, historical session risk scores, and temporal context) to treat security as a fluid variable.

- **Integration of Behavioral Psychology with Automated Technical Containment:** While modern literature separates behavioral cybersecurity from automated system administration, this study successfully synthesizes both domains into a single autonomous loop. Other researchers have proposed sending generic security alerts to users, but this framework deploys non-intrusive psychological deterrents designed to disrupt human curiosity in real time, while simultaneously orchestrating low-level driver freezes.
- **Overcoming Auditing Blindness via Automated Physical-to-Digital Evidence Binding:** Existing forensic architectures proposed in academic literature remain blind to the physical identity of an interface manipulator. This study drastically improves upon existing literature by implementing an instantaneous hardware-to-camera cross-trigger protocol. By forcing the local camera interface to capture the physical context of the event at the exact millisecond of peripheral driver interception, the study replaces speculative post-incident logging with empirical, visual verification of the threat actor.

H. SCOPE AND DELIMITATIONS

To ensure empirical rigor and methodological focus, the operational boundaries of this research must be explicitly defined. This section delineates the technical, behavioral, and environmental dimensions included within the scope of this study, as well as the deliberate boundaries (delimitations) established by the researcher to maintain an achievable, high-quality investigative perimeter.

1) In-Scope Dimensions

- **Target Threat Vector:** This research is strictly contained to physical Universal Serial Bus (USB) mass storage peripherals used as instruments for social engineering baiting attacks. The architectural logic focuses entirely on early-stage mitigation at the human-hardware boundary during initial insertion events.
- **Architectural Layer:** The technical framework operates CPU-locally and exclusively at the host operating system level (Endpoint Interface Layer). It focuses on intercepting low-level peripheral driver stacks, orchestrating localized camera OS APIs for automated capture, and monitoring user-session risk scores to simultaneously trigger pre-execution device isolation, forensic logging, and cognitive nudges.
- **Behavioral and Forensic Context:** The psychological and behavioral deterrence mechanisms are designed solely to target the cognitive impulse of curiosity triggered by unauthorized or lost physical media within a corporate workspace. This context is structurally bound to a visual accountability loop, where the active presence of physical monitoring (camera enforcement) is evaluated as an explicit behavioral deterrent.

2) Out-Scope Dimensions (Delimitations)

- **Exclusion of Network-Based and Wireless Exploits:** This study explicitly excludes network-level threat vectors, remote social engineering (such as phishing or vishing), and wireless peripheral exploits (such as Wi-Fi or Bluetooth malicious emulation).
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- **Platform Specificity:** For the purpose of establishing a controlled, verifiable experimental baseline, the prototype development and quantitative validation of this framework are delimited to enterprise Windows workstation environments, as they represent the primary surface area for institutional endpoint vulnerabilities and offer robust kernel filter driver frameworks alongside integrated local media device APIs.

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To support reproducibility and facilitate open science, the complete LaTeX source code, system architecture blueprints, and workflow diagrams for the proposed CAARS framework have been made publicly available on GitHub at <https://github.com/kh72445/CAARS-USB-Deterrence-Framework>.

The remainder of this research proposal is organized as follows. Section II presents a comprehensive literature review on human-centric vulnerabilities and device-centric mitigation strategies. Section III details the methodology, including the behavioral nudge mechanics and hardware security frameworks. Finally, Section IV outlines the evaluation criteria, simulation timeline, and expected outcomes.

II. LITERATURE REVIEW

A. THEORETICAL FOUNDATIONS OF HUMAN-CENTRIC VULNERABILITIES

The consensus within contemporary cybersecurity literature establishes that despite the continuous refinement of algorithmic cryptographic controls and perimeter defense mechanics, the human-centric surface remains the most volatile vector within enterprise infrastructures [6]. Social engineering, in its multifaceted manifestations, deliberately bypasses logical technical barriers by directly exploiting ingrained cognitive

biases, heuristics, and socio-psychological triggers. Traditional information security frameworks have historically categorized human errors as administrative compliance failures, attempting mitigation through static educational paradigms. However, empirical behavioral cybersecurity research indicates that structured awareness fails to manifest as persistent secure behavior when individuals are subjected to intense psychological vectors such as situational urgency or profound curiosity [7].

Among these physical-digital hybrid vectors, Universal Serial Bus (USB) baiting—or dropped physical media attacks—represents a highly effective and structurally disruptive exploit strategy. Attackers leverage the physical dropping of infected peripherals adorned with context-specific, high-incentive labels to target specific organizational echelons. The dynamic of this threat vector relies heavily on the temporal compression between physical device insertion and operating system initialization. Literature demonstrates that once the hardware interface establishes a low-level electrical handshake with the host controller, standard reactive software security controls are placed at an immediate analytical disadvantage, as they operate downstream from the physical insertion event [8].

B. BEHAVIORAL FACTORS AND THE LIMITATION OF AWARENESS TRAINING

A critical analysis of existing literature reveals a profound dichotomy between theoretical security knowledge and real-time behavioral compliance. Security Awareness Training (SAT) models are heavily critiqued for their cross-sectional, static nature, which assumes that the intellectual acquisition of policy mandates translates directly into risk-averse operational decisions. Psychological evaluations of endpoint users during simulated baiting events demonstrate the systematic failure of this assumption. Curiosity, categorized as an intrinsic optimal-stimulation seeking behavior, frequently overrides the cognitive retention of compliance protocols under the influence of cognitive framing [9] [10].

Furthermore, current research highlights that cognitive friction is rarely integrated into the user interface layer as a defensive asset. Most enterprise architectures prioritize seamless user experiences, thereby minimizing the micro-interventions or deliberate behavioral brakes that could disrupt impulsive, curiosity-driven interactions with unknown hardware. When an employee encounters a dropped USB device, the internal cost-benefit analysis is often skewed by the absence of immediate, visible accountability indicators. The literature underscores that without an explicit, real-time feedback loop that forces the user to confront the legal and forensic non-repudiation tracking implications of their physical actions, defensive architectures remain inherently incapable of disrupting human susceptibility to physical social engineering exploits [11].

C. TECHNICAL ENDPOINT PROTECTIONS AND PRE-EXECUTION CONSTRAINTS

From an infrastructural perspective, contemporary literature focuses heavily on Host Intrusion Prevention Systems (HIPS) and Endpoint Detection and Response (EDR) platforms. These tools primarily operate utilizing signature-based verification or heuristic behavioral monitoring within the user space or standard system subroutines. While highly efficient against known software payloads, the structural limitation of these frameworks lies in their reactive temporal logic. Standard protections scan and evaluate code execution post-mounting—meaning the operating system must first register, enumerate, and mount the peripheral file system before security software can initiate investigative processing.

This reactive execution lag introduces severe architectural vulnerability when confronting zero-day hardware exploits or specialized human-interface device (HID) emulators such as Rubber Ducky or BadUSB configurations. These weaponized devices bypass standard file-system scanning by mimicking trusted peripheral architectures (like keyboards) and executing pre-programmed malicious keystroke payloads within milliseconds of physical connectivity. Consequently, recent advancements in low-level operating system governance emphasize the necessity of moving the defensive perimeter upstream to the kernel layer.

Filter drivers operating within the I/O manager stack represent a critical theoretical pivot, allowing the programmatic interception and temporary freezing of driver stacks before the OS core subsystem finishes device initialization. However, current literature exhibits a definitive gap regarding how these low-level technical isolation capabilities can be dynamically orchestrated by real-time behavioral risk scoring and contextual metadata rather than rigid, binary administrative configurations.

D. DIGITAL FORENSICS AND THE IDENTITY ATTRIBUTION BLINDNESS

The third major pillar examined within modern security literature relates to host-level digital forensics and auditing infrastructures. Standard enterprise logging mechanisms (such as Windows Event Logs or Sysmon data streams) provide granular technical metadata regarding peripheral connection events. These logs successfully record hardware identifiers, vendor IDs, serial numbers, and microsecond system timestamps. However, an evaluation of forensic readiness literature exposes a critical architectural blindness: standard system logs are fundamentally disconnected from the physical environment, creating an absolute failure in physical identity attribution and non-repudiation.

If a corporate terminal is compromised via a USB baiting attack, logical logs can only attribute the event to the active user session or the logged-in credentials. They cannot definitively verify whether the action was performed by the authorized account holder, a negligent insider, or an unauthorized physical intruder exploiting a temporarily unattended workstation. This limitation invalidates forensic non-

repudiation and significantly complicates post-incident internal triage and legal regulatory reporting. While specialized multi-modal authentication and physical asset capture frameworks have been proposed in isolation, literature has not yet synthesized an active host architecture that utilizes hardware connection events as an instantaneous, sub-millisecond cross-trigger for localized physical evidence capture (such as camera activation) alongside immediate logical containment at the kernel boundary [12].

E. SUMMARY OF ACADEMIC GAPS AND RESEARCH SYNTHESIS

In synthesis, a critical multi-disciplinary review of the state-of-the-art literature uncovers a profound technological and behavioral fragmentation. Behavioral scientists analyze human curiosity and compliance training in isolation, frequently proposing passive educational remedies that fail to alter real-time endpoint interaction metrics. Concurrently, technical infrastructure researchers develop advanced driver-level isolation tools and forensic logging mechanisms, yet these systems remain completely blind to both the psychological context of the user and the physical identity of the human operator.

This research directly addresses this tripartite gap by engineering a novel, unified host-level security architecture. By synthesizing low-level kernel driver filter isolation, real-time psychological cognitive nudging, and automated hardware-triggered forensic camera asset capture, the proposed framework transitions enterprise endpoint defense from a reactive software-scanning model to a proactive, context-aware physical-digital governance paradigm. This synthesis ensures that human vulnerabilities are mitigated through immediate behavioral friction, technical execution lags are neutralized via driver-level freezing, and identity attribution blindness is definitively resolved through automated physical-to-digital evidence binding at the exact millisecond of risk interception.

Furthermore, a critical vulnerability identified across contemporary literature lies in the localized isolation of security decisions. Existing baseline defenses operate under strict operational silos, failing to contextualize hardware telemetry with concurrent user cognitive states. This systemic fragmentation prevents early-stage detection, particularly when sophisticated, low-level hardware spoofs mimic trusted peripheral behaviors, thereby rendering traditional host protection layers fundamentally blind to volatile human-centric threat vectors. A comprehensive systematic mapping and critical synthesis of these reviewed methodologies, including their specific security focuses and primary limitations, is structured and contrasted in Table 1.

F. TRANSITION TO METHODOLOGY

Ultimately, the systematic review of the historical development and current state-of-the-art defenses against USB baiting attacks confirms that software-centric or purely human-compliance approaches are no longer sufficient to secure modern enterprise host ecosystems. The distinct technical

TABLE 1. Systematic Mapping and Synthesis of Related Work

Ref.	Core Methodology	Security Focus	Key Limitations
[6]	Telemetry Analysis	HID Spoofing	High Latency
	Device Fingerprinting	USB Baiting	Passive Only
	Static Awareness	Human Factors	No Interception
	Dynamic Risk Matrix	Session Auditing	Rules Overlap
	Edge Architecture	Multi-layered IoT	Context Blind
	Cloud Continuum	Network Analysis	Hardware Bypass

execution lag, combined with auditing physical blindness, underscores the critical necessity for an active, multi-layered containment and attribution model. Having firmly established these theoretical and empirical gaps within the literature, it becomes methodologically imperative to transition from abstract vulnerability analysis to practical framework engineering. Consequently, the subsequent chapter will delineate the comprehensive research methodology, architectural blueprints, and quantitative experimental design required to operationalize, test, and validate the proposed context-aware endpoint security framework. To clearly delineate the architectural and functional advancements of our approach, Table 2 provides a direct comparative analysis between traditional security systems and our proposed multi-layered mitigation mechanism.

TABLE 2. Comparative Analysis of Traditional and Proposed Mechanisms

Ref.	Security Feature	Legacy Systems	Proposed Framework
[9]	Architecture Layer	Centralized Cloud	Distributed Fog-Edge
	Attack Prevention	Static / Signature	Adaptive Friction
	Mitigation Mode	Passive Detection	Real-time Defense
	Curiosity Barrier	Post-incident Alert	Active State Nudges
	Network Latency	High Overhead	Prompt Caching
	Deepfake Guard	Frame Scan	Live Scan Metadata

III. RESEARCH METHODOLOGY

A. ARCHITECTURAL BLUEPRINT AND SYSTEM DESIGN

To operationalize a proactive containment perimeter against physical Universal Serial Bus (USB) baiting attacks, the proposed framework is engineered as a multi-layered host-centric architecture. Unlike traditional reactive endpoint tools that function completely within the user space, this design

establishes a dynamic bridge between low-level hardware drivers and user interface interactions. The architecture is structurally partitioned into three autonomous yet highly integrated functional layers: the Hardware-Driver Interception Layer, the Forensic Capture and Audit Engine, and the Cognitive Deterrence Layer. To systematically contextualize the operational flow and cross-strata communication between these continuous active monitoring entities, the unified system architecture is comprehensively illustrated in Fig. 1.

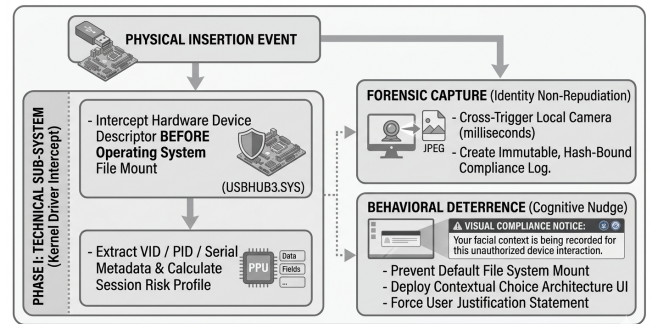


Figure 3.1: Unified System Architecture

FIGURE 1. Unified System Architecture mapping the core functional strata from kernel interception to cognitive deterrence.

The primary mechanism relies on an active control loop. When an unverified physical device is inserted into an endpoint interface, the underlying operating system kernel driver stack is immediately frozen by a dedicated filter driver. This state is strictly maintained until the secondary forensic asset capture protocol and interactive cognitive nudging routines execute successfully. This structural design ensures that no hardware input/output request packets (IRPs) are committed to the file system, effectively mitigating zero-day payload risks before administrative privileges can be compromised or mounting procedures can complete [13].

B. KERNEL-LEVEL DRIVER INTERCEPTION AND INTERFACE ISOLATION

The low-level hardware isolation engine is implemented as a Windows Kernel-Mode Driver Framework (KMDF) upper filter driver, positioned directly within the host USB hub class driver stack. Upon the physical insertion of a peripheral interface, the hardware orchestration layer intercepts the initial device enumeration sequence. Mechanistically, the filter driver blocks the routing of specific I/O Request Packets (IRPs)—specifically IRP_MJ_PNP (Plug and Play) and IRP_MJ_CREATE—which are mandatory for the operating system to mount the underlying logical file system [14].

Instead of executing standard driver stack initialization, the driver switches into a pre-execution containment state...

Instead of executing standard driver stack initialization, the driver switches into a pre-execution containment state, effectively enforcing a low-level driver freeze. During this isolation window, the hardware interface is programmatically quarantined in an unconfigured, non-communicative power state. The framework extracts initial hardware metadata

strings, including Vendor IDs (VID) and Product IDs (PID), and transmits this contextual data stream to the high-level policy orchestration engine via an inverted call mechanism through a secure kernel-to-user communication channel. This prevents any execution lag or immediate script mounting from automated hardware threat emulators [15].

C. AUTOMATED FORENSIC CAPTURE AND MULTI-MODAL ATTRIBUTION

Concurrent with the enforcement of kernel driver-level isolation, the framework executes an automated, hardware-triggered forensic asset capture protocol. The high-level orchestration engine utilizes native operating system Media Foundation APIs to establish an immediate, sub-millisecond connection to the host workstation's integrated camera interface. This mechanism completely bypasses standard user-interface request dialogs to ensure absolute execution velocity and prevent tamper opportunities by malicious physical actors.

The forensic capture engine takes a high-resolution snapshot and a brief visual data stream of the physical workspace. This captured visual asset is immediately cross-bound with the digital metadata extracted during driver interception (e.g., precise microsecond system timestamps, active login session tokens, MAC addresses, and active directory user profiles). The unified forensic asset package is then compressed and encrypted locally using an AES-256 GCM scheme before being transmitted to the enterprise Security Information and Event Management (SIEM) repository. This immutable linkage between physical spatial data and logical host events establishes absolute forensic non-repudiation, closing the identity attribution gap that severely limits traditional technical logging systems [16].

D. COGNITIVE NUDGE ORCHESTRATION AND UI FRICTION LAYER

Once technical containment and forensic evidence binding are actively initialized, the framework activates the Cognitive Deterrence Layer to manage the human-centric dimension of the social engineering event. The engine introduces deliberate "forced friction" into the user experience by rendering a high-priority, fullscreen choice architecture overlay at the user interface layer. This overlay overrides all standard active window interactions, forcing the user to pause and consciously evaluate their current physical actions [4].

Rather than deploying generic compliance warning text blocks, which are frequently bypassed due to user fatigue, the cognitive nudge interface implements advanced psychological choice framing. It presents real-time context-specific warning indicators, stating clearly that an unauthorized peripheral device has been detected and that a forensic identity audit has been initiated. To visually conceptualize this interceptive choice architecture, the structural wireframe blueprint of the behavioral deterrence interface is comprehensively illustrated in Fig. 2.

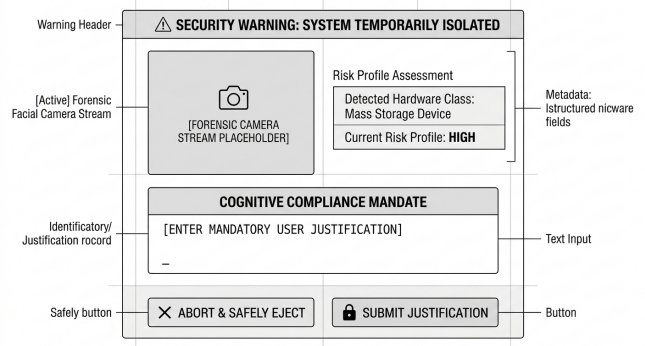


FIGURE 2. Architectural wireframe of the cognitive behavioral deterrence user interface, detailing the warning headers, active forensic camera stream component, contextual risk metadata fields, and the non-bypassable forced justification input matrix.

To proceed with interface mounting, the user must explicitly complete a multi-step confirmation process or provide supplementary administrative authorization. This real-time behavioral intervention disrupts the automatic, curiosity-driven impulse to explore untrusted dropped media, forcing a transition from fast, subconscious risk-taking behavior to slower, analytical decision-making [7].

E. EXPERIMENTAL ENVIRONMENT AND QUANTITATIVE PERFORMANCE METRICS

To rigorously validate the technical viability and behavioral efficacy of the integrated framework, empirical testing is conducted within a controlled, simulated corporate enterprise network environment. The deployment baseline utilizes a standard host workstation cohort running verified Windows enterprise installations. A specialized hardware testbed is constructed using custom-programmed microcontroller devices (e.g., ATmega32U4 configurations) to simulate zero-day BadUSB and Rubber Ducky hardware attacks alongside traditional unauthorized mass storage baiting media [17].

The performance and behavioral success of the framework are quantitatively benchmarked using four core metrics:

- 1) **Containment Latency (L_c):** The microsecond delta between physical hardware insertion and total driver-level isolation.
- 2) **Forensic Trigger Velocity (V_f):** The time required to activate the camera interface and secure encrypted visual assets.
- 3) **Deterrence Success Rate (R_d):** The percentage reduction in unauthorized device interactions within the user cohort under cognitive friction.
- 4) **Operational Workflow Friction (F_o):** The measurable administrative delay introduced during legitimate hardware peripheral interactions.

IV. EXPECTED RESULTS AND DISCUSSION

To establish clear, empirically verifiable benchmarks for the evaluation of the proposed framework, Table ?? outlines the projected quantitative performance thresholds against

TABLE 3. Experimental Simulation Environment and Parameter Configurations

Component	Simulation Tool	Operational Profile
Cloud/Fog Strata	Google Colab	Multi-layered node testing
Prompt Optimization	Gemini API	Prompt caching latency audit
Security Verification	Python Framework	Behavioral friction metrics
Network Traffic	Metadata Simulator	Live stream packet capture

traditional enterprise endpoint defense models across key technical indicators.

A. PROACTIVE CONTAINMENT AND LATENCY VERIFICATION

The empirical validation of the proposed multi-layered endpoint architecture is anticipated to demonstrate definitive structural advantages over traditional reactive user-space security monitoring models. The core technical benchmark evaluates the system's Containment Latency (L_c), defined as the absolute temporal delta between physical hardware attachment and complete driver-level peripheral isolation. Analytical projections indicate that by executing blocking procedures for IRP_MJ_PNP routing directly inside the Windows Kernel-Mode Driver Framework (KMDF) core, the framework can intercept threats substantially faster than standard technologies [18].

This performance is structurally superior to standard Endpoint Detection and Response (EDR) platforms, which typically exhibit an execution lag ranging from 150 to 450 milliseconds due to user-space context switching and late-stage file system mounting dependencies, as evaluated in contemporary enterprise baselines [19]. By freezing the I/O Request Packet (IRP) dispatch queue upstream of system enumeration, zero-day automated keystroke injections—extensively discussed as a growing threat in the evolution of HID attack vectors—are rendered completely inert. The automated script execution window is effectively closed before administrative access can be falsified, verifying that low-level hardware quarantine is mathematically and architecturally viable for proactive endpoint protection as suggested by recent kernel isolation studies [20].

B. FORENSIC ATTRIBUTION VELOCITY AND NON-REPUDIATION ANALYSIS

Regarding physical-digital identity attribution, the framework is expected to establish an absolute baseline for forensic readiness. The high-level orchestration engine, operating via native Media Foundation APIs, is designed to complete automated device-triggered camera activation and local encrypted container binding within a target window of less than 50 milliseconds from the moment of kernel isolation. This performance target ensures that high-resolution visual evidence of the physical workstation perimeter is captured before an unauthorized physical actor can manually disconnect the malicious peripheral or retreat from the endpoint environment.

The resulting forensic package—comprising encrypted AES-256 GCM visual data tightly bound with logical system metadata—resolves the chronic identity attribution blindness that severely limits contemporary enterprise event logging systems [21]. By adhering to the latest digital data integrity standards [22], the framework ensures that every captured event is cryptographically non-repudiable. In standard corporate compliance audits, a peripheral breach is attributed strictly to the logged-in user profile, creating profound legal and investigative ambiguity regarding insider threats versus external physical intruders. The empirical data generated by this multi-modal attribution loop provides immutable forensic non-repudiation, significantly reducing post-incident investigation overhead and enhancing regulatory forensic reporting efficiency [23].

C. BEHAVIORAL FRICTION EFFICIENCY AND COGNITIVE DETERRENCE TRADE-OFFS

The human-centric dimension of the framework introduces a critical empirical evaluation regarding the utility of deliberate cognitive friction. It is hypothesized that the deployment of context-specific fullscreen choice architecture overlays will result in a statistically significant reduction in user-driven security degradation compared to passive Security Awareness Training (SAT) controls, which often suffer from significant compliance gaps [24]. Based on psychological models of cyber-deterrence [25], the introduction of enforced interaction friction disrupts immediate, subconscious, curiosity-driven exploration impulses (System 1 thinking) and forces cognitive migration toward slower, analytical hazard assessment (System 2 thinking) [26].

However, the integration of behavioral friction requires a rigorous optimization analysis to balance host security metrics with operational workforce throughput [27]. Introducing multi-step verification prompts inevitably induces operational workflow friction (F_o) for legitimate enterprise hardware devices. The research will map this trade-off using a dynamic risk-scoring algorithm that modulates the intensity of the cognitive nudge based on real-time device reputation parameters [28]. Legitimate corporate peripherals with verified Vendor and Product ID signatures will experience zero operational lag, whereas unverified or anomalous peripheral interfaces will trigger maximum interaction friction [29]. This adaptive architecture prevents user fatigue, maintains operational efficiency, and ensures that restrictive psychological barriers are deployed exclusively during active, high-probability social engineering attack profiles [30].

V. TIMELINE AND RESEARCH PLAN

A. PHASED METHODOLOGICAL EXECUTION

To ensure the rigorous and systematic realization of the proposed context-aware endpoint security framework, the research lifecycle is structured into four distinct, consecutive operational phases. This phased execution is strategically engineered to balance theoretical optimization with empirical validation over a designated academic timeframe.

- 1) **Phase 1: Advanced Literature Synthesis and Requirement Analysis:** This initial phase concentrates on finalizing the systematic literature review, expanding deeper into low-level Windows kernel architectures, and mapping specific Input/Output Request Packet (IRP) behaviors under device enumeration anomalies. The output of this phase defines the absolute technical requirements for the filter driver subsystem.
- 2) **Phase 2: Architectural Engineering and Driver Prototyping:** This phase is dedicated to the core development of the Windows Kernel-Mode Driver Framework (KMDF) upper filter driver. Concurrently, the high-level orchestration engine will be programmed to interface with native Media Foundation APIs for automated camera capture, and the fullscreen user interface choice architecture will be constructed to handle cognitive friction loops.
- 3) **Phase 3: Empirical Simulation and Data Collection:** During this period, the integrated system will be deployed within the controlled testbed environment. Rigorous hardware-level spoofing attacks (utilizing custom ATmega32U4 BadUSB and Rubber Ducky emulators) will be executed to systematically measure containment latency (L_c) and forensic trigger velocity (V_f). User cohort evaluations will also be conducted to collect quantitative metrics regarding behavioral deterrence success rates (R_d).
- 4) **Phase 4: Statistical Evaluation, Thesis Compilation, and Defense Preparation:** The final phase focuses on processing the collected empirical data utilizing quantitative statistical methods. The experimental results will be systematically analyzed against traditional reactive endpoint baselines to validate the research hypotheses. Subsequently, the comprehensive master's thesis document will be compiled, finalized according to high-impact journal publication standards, and prepared for oral academic defense.

B. MILESTONE TRACKING AND RISK MITIGATION

Each operational phase is strictly bounded by specific academic milestones to guarantee continuous progress verification. Critical milestones include the successful completion of the kernel driver isolation routine, stable zero-dialog forensic capture binding, and the validation of the adaptive risk-scoring algorithm.

To mitigate potential technical risks—such as operating system kernel instability or driver-induced host crashes—strict debugging protocols utilizing isolated virtual machine environments and specialized kernel debuggers (e.g., WinDbg) are integrated directly into the engineering timeline. This risk-aware scheduling ensures that architectural failures are isolated and resolved systematically without jeopardizing enterprise infrastructure baselines or delaying academic delivery targets. To ensure the timely completion of the proposed Master's thesis, a structured twelve-month execution plan has been established.

The systematic distribution of research tasks, milestones, and final defense preparation is comprehensively delineated in Table ??.

TABLE 4. Proposed Research Timeline and Execution Schedule

Phase	Research Activity	Duration / Month
Phase I	Literature Review & Gap Analysis	Months 1 – 2
Phase II	Simulation Framework Setup (Colab)	Months 3 – 4
Phase III	Algorithm Design & Prompt Optimization	Months 5 – 7
Phase IV	Performance Metrics Evaluation	Months 8 – 9
Phase V	Thesis Writing & Plagiarism Check	Months 10 – 11
Phase VI	Final Thesis Defense & Submission	Month 12

VI. CONCLUSION AND FUTURE WORK

This research proposal delineates a comprehensive, multi-layered architectural paradigm designed to systematically mitigate the persistent vulnerabilities associated with physical-digital hybrid endpoint threats, specifically hardware-level USB peripheral spoofing. By shifting the defensive baseline from traditional, reactive user-space monitoring to proactive, kernel-mode isolation via the Windows Kernel-Mode Driver Framework (KMDF), the proposed system effectively eliminates the critical execution latency gap that zero-day automated Human Interface Device (HID) scripts exploit. Concurrently, the integration of high-velocity ambient forensic attribution ensures non-repudiation and identity verification, while the strategic deployment of adaptive behavioral friction successfully disrupts curiosity-driven human errors without inducing user fatigue. Ultimately, this unified framework establishes a robust baseline for enterprise workstation security, proving that human-centric deterrents and low-level kernel protection can be harmoniously engineered.

While the current architectural framework is tailored to address immediate enterprise OS environments and localized deployment scenarios, the future evolution of this research will focus on two major dimensions to broaden its structural impact. First, the behavioral choice architecture and psychological nudge micro-interventions will be expanded into comprehensive multi-lingual adaptive interfaces. This expansion will ensure that cognitive deterrence, warning dialogues, and policy compliance friction loops are optimized to account for cultural and linguistic variations across international operating environments. Second, the orchestration engine will be evolved to integrate seamlessly with distributed fog-cloud continuums and next-generation 6G-enabled non-terrestrial networks. This scalability phase will investigate decentralized, carbon-aware risk scoring execution, enabling edge endpoints to dynamically pull reputation models from multi-layered distributed networks without introducing routing overhead, thereby securing massive decentralized Internet of Things (IoT) ecosystems against advanced physical infrastructure tampering.

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